

Philosophy of Space and Time

Lecture 12: *Measurement, Geometry, and the Block Universe*

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Today's Plan

- 1 Recap
- 2 What Clocks and Rods Measure
- 3 Rods and Clocks
- 4 Geometry or Dynamics?
- 5 The Block Universe
- 6 Where We Now Stand
- 7 Closing

Where We Left Off

Last time we developed the **geometry of Minkowski spacetime**:

- The **invariant interval** $I = t_1 t_2$ (we will revisit Geroch's construction)
- **Light cones** replace simultaneity slices as the basic structure, three kinds of separation: timelike, spacelike, null
- **Twin paradox**: inertial worldlines maximize proper time

Two Philosophical Questions

Question 1: The role of spacetime geometry?

The interval is the geometric quantity, access to it based on behavior of physical clocks and rods. Is the geometry then *explanatory*, or is it just a summary of how matter behaves?

Question 2: What is left of “the present”?

With absolute simultaneity gone, is there still an objective “now”? How should we think about the nature of “the present” and the passage of time?

Both questions concern the *interpretation* of the geometric structure we have introduced.

Spacetime Geometry

The Operational Reconstruction

Geroch's strategy (Chs. 5–6): build the geometry from the bottom up, using only

- **ideal clocks** (carried along worldlines)
- **light signals** (emitted, reflected, received)

From these two primitives we derived:

- the interval between any two events
- the timelike / spacelike / null distinction
- the light-cone structure

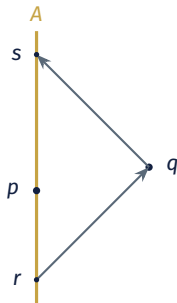
No appeal to rigid measuring rods, because they are too complex; only to clocks plus light.

Interpreting the Interval

Last time we defined the interval in terms of A 's clock readings t_1, t_2 . We can also interpret what A would say about the distance between p and q (Geroch §5.6, pp. 82–84):

- What is the distance between p and q ?
- Light ray goes to q with a total round-trip time: $t = t_1 + t_2$. Since q is at the half-way point, the distance is then:

$$\Delta x = \frac{c}{2}(t_1 + t_2)$$



Interpreting the Interval, 2

What is the time elapsed between p and q ?

- Find p' that is simultaneous with q , compare p and p' .
- Total elapsed time between r and s

$$t_{\text{Total}} = (t_1 + t_2)$$

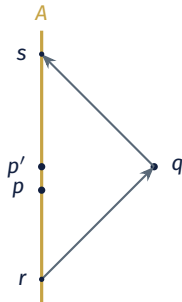
- p' is halfway between r, s , or

$$\frac{1}{2}(t_1 + t_2)$$

after r . And p is just t_2 after r .

- Time difference is then

$$\begin{aligned}\Delta t &= \frac{1}{2}(t_1 + t_2) - t_2 \\ &= \frac{1}{2}(t_1 - t_2)\end{aligned}$$



Interpreting the Interval, 3

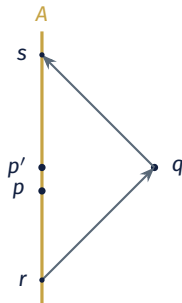
So overall the “apparent distance and time” between p, q in terms of coordinates are:

- Distance in time:

$$\Delta t = \frac{1}{2}(t_1 - t_2)$$

- Spatial distance

$$\Delta x = \frac{c}{2}(t_1 + t_2)$$



The Interval in Coordinates

Inverting the relations (Geroch, pp. 96-97):

$$t_1 = \frac{\Delta x}{c} + \Delta t, \quad t_2 = \frac{\Delta x}{c} - \Delta t.$$

So then the interval $I = t_1 t_2$ is given by:

$$I(p, q) = \left(\frac{\Delta x}{c} + \Delta t\right) \left(\frac{\Delta x}{c} - \Delta t\right) = \frac{(\Delta x)^2}{c^2} - (\Delta t)^2.$$

- Different inertial observers assign different t_1, t_2 , hence different $\Delta t, \Delta x$
- But every observer computes the *same product*

A Minkowski “Compass” (Maudlin pp. 85–87)

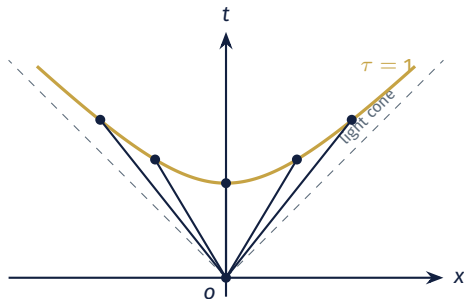
Maudlin’s geometric analog of a Euclidean compass:

- Identical alarm clocks, all set to ring after 1 minute of proper time
- All sent inertially from event o , with different velocities and directions

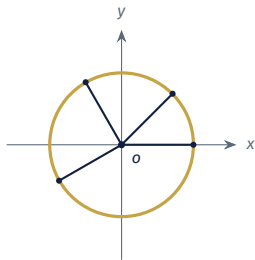
Locus of ringing events:

$$t^2 - x^2 - y^2 - z^2 = 1.$$

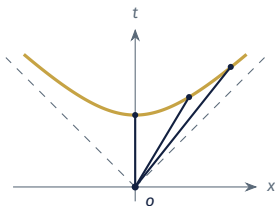
In a 1+1 diagram, a **hyperbola**; in 3+1, a **hyperboloid of revolution**.



The Hyperbola Is the Minkowski “Circle”



Euclidean
 $x^2 + y^2 = r^2$



Minkowski
 $t^2 - x^2 = \tau^2$

- Each curve is the locus of points at the same intrinsic “distance” from o
- Asymptotes = the light cone: null trajectories have proper time zero
- **Time dilation**: a fast-moving clock travels far in x before its alarm rings

Rods and Clocks in Minkowski Spacetime

The Set-up: Beth and Paul

A pole approaches a barn at high relativistic speed. The pole's rest length is greater than the barn's rest length; the barn has open doors at both its near and far ends.

- **Beth**: at rest in the barn's frame
- **Paul**: at rest in the pole's frame
- In Beth's frame, the pole is Lorentz-contracted to (essentially) the barn's length

Two events define the puzzle:

- **Event A**: the front of the pole is at the far door
- **Event B**: the back of the pole is at the near door

Beth and Paul agree on which events occur — but disagree on their temporal order.

Two Frames, Two Verdicts

Beth (barn's frame):

- The contracted pole (just) fits the barn
- *A and B are simultaneous*
- “At one moment, both ends of the pole are at their respective doors”
- The pole *fits* inside the barn

Paul (pole's frame):

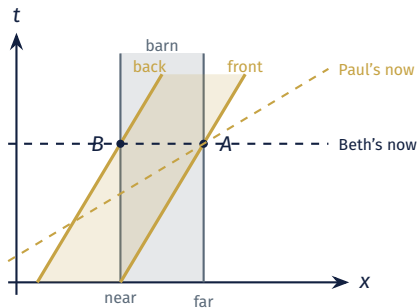
- The barn is contracted; shorter than his (rest-length) pole
- *A occurs before B*
- “The front exits the far door before the back reaches the near door”
- The pole *never* fits

?? *Both descriptions are correct in their own frame. Does the pole fit in the barn or not?*

The Spacetime Picture

- Pole's worldsheet: tilted strip
- Barn's worldsheet: vertical strip
- Beth's slice through A also passes through B — they are simultaneous for her
- Paul's slice through A does *not* pass through B — B lies later on his slicing

Each observer takes a different cross-section of the same pole worldsheet.



The Lesson

- Length contraction is not a perceptual illusion — the contracted length is what one really measures by the simultaneous-endpoint procedure
- But it is not a frame-independent fact about the rod — it is a fact about the rod *as sliced* by a particular surface of simultaneity
- The geometric quantity (the rod's worldsheet, with its rest length) is the same in every frame

Maudlin's framing

Length contraction and time dilation are real, but they are also perspectival: they result from *relations* between a rod or clock and a frame's slicing.

Several Ways to Measure Length

Geroch (§5.7) considers three distinct procedures for finding the spatial length of a moving rod:

1. **Simultaneous-endpoint method**: mark the positions of both ends “at the same time” in some inertial frame
2. **Radar method**: bounce a light pulse from one end to the other and back; multiply round-trip time by $c/2$
3. **Comoving rest-length**: the length the rod has in its own rest frame

In Newtonian physics these all give the same answer. In relativity they typically do *not* — but the discrepancies are calculable from the interval.

Geroch's point

The various length-measurement procedures give different numbers, but the differences are systematic and frame-relative. Each procedure is *compatible* with the others, given the interval structure.

- No single procedure is privileged
- “Length” in relativity is a frame-relative quantity, derived from the invariant interval
- The same is true for “elapsed time” — it is proper time along a specified worldline

Compare Newtonian physics, where length and elapsed time were assumed to be observer-independent.

**Geometry
or Dynamics?**

Two Readings of Special Relativity

Geometric (Minkowski, Maudlin)

The geometry of spacetime is fundamental. The behavior of rods and clocks is *explained by* their motion through that geometry.

Dynamical (Lorentz, Brown)

The Lorentz-invariant dynamics of matter is fundamental. The Minkowski metric is just a *summary* of how matter happens to behave.

Time dilation and length contraction are not perceptual artifacts or measurement quirks; they are direct consequences of the geometry of Minkowski spacetime.

- Spacetime has the Minkowski metric as a real geometric structure, and this *constrains* what physical systems can do
- Any method of measuring length or time will exhibit relativistic effects

The interval is a sort of misty thing that stands in the background, dominating all that goes on (for it is the only link we have between the observations of different individuals), but disappearing in the end. It's a beautiful idea – and a rather subtle one – having the crucial quantity be something which doesn't by itself have all that much physical significance. Rather, its significance comes from its implications in these and other similar calculations. ... One “understands” the interval only through its physical implications. But what more could one ask for as a constituent of “understanding”?

— Geroch, p. 158

Brown: Dynamics as Fundamental

Harvey Brown (*Physical Relativity*, 2005) emphasizes the dynamics of matter theory.

- Rods and clocks are made of matter
- Their behavior under boosts is determined by the dynamical laws governing their constituents (electromagnetism, the strong force, ...)
- Those laws are Lorentz invariant
- Length contraction and time dilation are *consequences* of this dynamical Lorentz invariance

The Minkowski metric is no more than a codification of the Lorentz-invariant behavior of rods and clocks.

Brown's Criticism of Geometric View

On Brown's view, the geometric reading inverts the true order of explanation.

- Saying “the rod contracts because of the spacetime geometry” is like saying “the planets follow ellipses because they follow grooves in spacetime”
- The geometry is descriptive, not explanatory
- The Lorentz invariance of dynamical laws is doing the real work

The Stakes

Maudlin and Brown agree on empirical consequences of relativity theory, but differ on interpretation and explanation:

- It is a debate about the *order of explanation*
- Does spacetime have an autonomous geometric structure that constrains matter?
- Or is “spacetime geometry” just a compact way of describing the symmetries of dynamical laws?

?? *What are the relevant considerations that can help to resolve this debate?*

The Block Universe

What Happened to “Now”?

Pre-relativistic physics had a clear “now”: the present is the global hyperplane $t = \text{now}$, in some absolute time.

- Past, present, future were absolute (shared by all observers)
- Pre-relativistic metaphysics could draw a sharp line between what *is* and what *has been* or *will be*

In Minkowski spacetime, there is no global “now” — different inertial observers slice spacetime into different surfaces of simultaneity.

?? *What happens to the distinction between past, present, and future when “the present” is no longer absolute?*

Putnam's Argument (1967)

Hilary Putnam offered a striking argument from the relativity of simultaneity to a **block universe** in which all events are equally real.

1. I am here and now (event E). Things real to me at E are real, full stop.
2. Another observer, momentarily co-located with me at E but in motion, has a different simultaneity slice through E
3. By the principle of relativity, things real to her at E are also real, full stop
4. By transitivity of “real”, everything on her slice is real
5. Iterate: every event in spacetime is real

Putnam's Conclusion

I conclude that the problem of the reality and the determinateness of future events is now solved.

— Putnam, “Time and Physical Geometry” (1967)

- Past, present, and future events are all equally real
- There is no objective “flow” of time, no privileged “now”
- The universe is a **block** — a four-dimensional manifold of events, all on the same ontological footing

Stein's Reply (1968, 1991)

Howard Stein argued that Putnam's argument depends on an unwarranted transitivity assumption.

- “Reality from the standpoint of E ” is naturally identified with E 's causal past — events that could have influenced E
- This relation is *not* transitive across spacelike-related observers
- In SR, the natural notion of “the determined past” relativizes to a worldpoint, not to a hyperplane
- So Putnam's premise (2)–(3) cannot be combined as he assumes

Stein's point: SR does not force a block universe; it forces us to reconceive what “the present” means.

The Two Pictures Compared

Putnam: Block universe

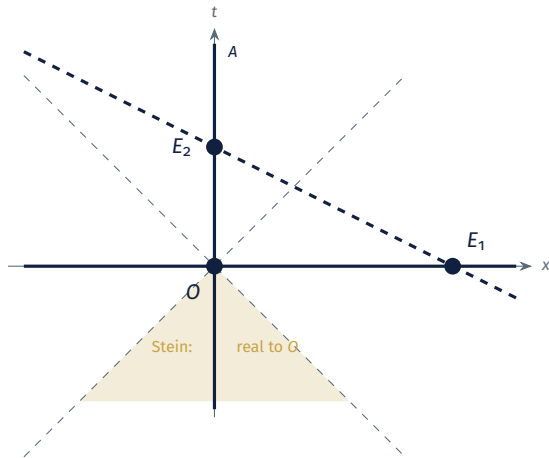
- All spacetime events equally real
- No objective passage
- “Becoming” is illusory

Stein: Worldpoint-relative

- “Real from E ” = E ’s past light cone
- No global “now”, but a local one
- Becoming is preserved, suitably reconceived

Both positions accept SR. They disagree on what SR implies for our notion of the “present,” and “becoming”.

Putnam vs. Stein: Two Geometries of “Now”



Putnam: *real-to- O* extends via transitivity.

- $O \sim E_1$ (in A 's frame)
- $E_1 \sim E_2$ (in C 's frame)
- Transitivity $\Rightarrow E_2$ real to O
- But E_2 lies in A 's future
- Iterating: every event is real to O

Stein: only the past light cone of O is determinately real to O .

- “Reality relation” respects causal structure
- No transitivity

Rovelli: “‘Now’ Means Nothing”

Rovelli's view undermines Putnam's position (*The Order of Time*, pp. 41–56):

- Putnam asks: *which events are real to O?*
- This presupposes a meaningful universal “now”—a global slice of simultaneity
- Relativity shows that we have a *partial order* rather than a *total order*

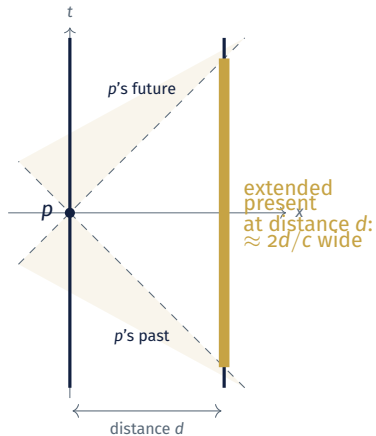
The (Partial) Order of Time

- Ask, “What is happening on **Proxima Centauri** right now?”
(Proxima is ~ 4 light-years away)
- An ~ 8 -year span of events on Proxima’s worldline is spacelike-separated from us here-now (these events are neither past nor future)
- Within that span, no event is privileged as “the present”

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- Within that span, no event is privileged as “the present”
- **The question has no determinate answer**

The Extended Present



The spacelike-separated region intersects a distant worldline at distance d in a band of width $\sim 2d/c$ — the **extended present**.

Concrete examples:

- Mars (avg. ~ 12 light-min): ~ 24 min
- Proxima Centauri (~ 4 ly): ~ 8 years
- Andromeda (~ 2.5 Mly): ~ 5 million years

Within each band, no event is the “now” of p . Locally, the band is so narrow that our intuitive “now” works fine.

Rovelli's Diagnosis

How does Rovelli's view bear on Putnam vs. Stein? What relativity leaves us with regarding temporal order:

- The light-cone structure (causal order)
- Proper time along worldlines (intrinsic geometry)
- Local temporal relations

?? *Presentism and eternalism share a faulty premise: that there is a global “now,” and then we can argue over its status. Relativity provides only local time – a partial order, not a total order.*

**Where We
Now Stand**

The Elimination Story Revisited

The progression we have built up over the last several lectures:

1. **Full Newtonian** spacetime: persistence of spatial points, absolute simultaneity, Euclidean space, inertial structure
2. **Galilean** spacetime: drop persistence of points (Leibniz's shift, Maudlin's mismatch)
3. **Minkowski** spacetime: drop absolute simultaneity (Einstein, light)

Each step removes structure that the dynamics did not require, and that was unobservable in the relevant sense.

What's Still There

Minkowski spacetime is not empty of structure. It still has:

- A **flat** metric, fixed once and for all
- A privileged class of inertial worldlines
- Light cones at every event
- A symmetry group: the Poincaré group

?? *Is this structure also excessive? Newton's spacetime had structure that classical dynamics did not require. Does Minkowski spacetime have structure that relativistic dynamics does not require?*

A Hint from Gravitation

The answer, when we get to general relativity, will be: yes, in a striking way.

- In Minkowski spacetime, the metric is a fixed background — it is not affected by what matter does
- This is in tension with a deep symmetry between gravitational and inertial mass that has been known since Galileo
- Einstein will conclude: the metric itself must be *dynamical* — it must respond to matter
- The fixed background of Minkowski spacetime is the next layer of excess structure to fall

This is the topic of the next two weeks.

Summary

1. **Rods and Clocks:** Geroch's operational reconstruction shows that several distinct procedures (radar, simultaneity, comoving) yield mutually compatible length-and-time data, all derivable from the interval
2. **Geometry vs. dynamics:** Maudlin treats the geometry as explanatory; Brown treats it as a summary of dynamical Lorentz invariance — conflicting interpretations
3. **Block universe:** Putnam argues that the relativity of simultaneity forces eternalism; Stein replies that becoming can be relativized to worldpoints
4. **What's next:** Minkowski's flat background metric is itself the next candidate for elimination

For Next Time

- **Read:** Maudlin, Ch. 6; Geroch, Chs. 7–8.
- **Think about:**
 - › Why is the equality of inertial and gravitational mass surprising in Newtonian physics?
 - › What does it mean to say that gravity is *not* a force?
 - › How could the metric of spacetime fail to be fixed once and for all?

May 14: **General Relativity I — The Equivalence Principle**

- Inertial vs. gravitational mass
- Einstein's happiest thought
- Tidal forces and the local-global distinction
- Free fall as inertial motion

The Clock Hypothesis (recap)

An **ideal clock** measures the proper-time integral along its worldline:

- This is the **clock hypothesis**: clocks are sensitive to the interval, not to the history of the clock or other factors
- Empirically very well confirmed (atomic clocks, muon decay, GPS satellites)
- But strictly speaking it is a *contingent* claim about the dynamics of physical clocks

Example: Muon Decay

Cosmic-ray muons give an everyday confirmation of the clock hypothesis (Geroch §5.5, p. 77):

- Created in the upper atmosphere (~ 15 km altitude); rest-frame lifetime $\tau_0 \approx 2.2 \mu\text{s}$
- Without time dilation, at $0.99c$ they would travel only ~ 650 m before decaying
- Observed: large numbers reach the ground

Two descriptions of the same fact:

- *Lab frame*: the muon's decay clock is dilated, so $\sim 2.2 \mu\text{s}$ of proper time stretches over ~ 15 km of lab travel
- *Muon frame*: the atmosphere is Lorentz-contracted to ~ 650 m, traversed in $\sim 2.2 \mu\text{s}$

Both descriptions agree on the proper time elapsed along the muon's worldline, and decay statistics depend on this proper time.

Questions?

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